



Published in final edited form as:

Clin Cancer Res. 2025 November 03; 31(21): 4405–4411. doi:10.1158/1078-0432.CCR-25-1388.

FDA Approval Summary: Datopotamab deruxtecan-dlnk for treatment of patients with unresectable or metastatic, HR-positive, HER2-negative breast cancer

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Abstract

On January 17, 2025, the FDA approved datopotamab deruxtecan-dlnk (DATROWAY, Dato-DXd), a Trop-2-directed antibody and topoisomerase inhibitor conjugate, for the treatment of adults with unresectable or metastatic, hormone receptor-positive (HR+), human epidermal growth factor receptor 2-negative (HER2-) breast cancer who have received prior endocrine-based therapy and chemotherapy for unresectable or metastatic disease. Approval was based on results from TROPION-Breast01 (TB01), a multicenter, randomized, open-label trial comparing Dato-DXd to investigator's choice of chemotherapy (ICC). The trial was designed with dual primary endpoints: progression-free survival (PFS) assessed by blinded independent central review (BICR) according to Response Evaluation Criteria in Solid Tumors (RECIST) v1.1 and overall survival (OS). TB01 demonstrated a 2-month improvement in median PFS for Dato-DXd compared to ICC (6.9 months vs. 4.9 months, respectively; stratified hazard ratio (HR) 0.63 (95% CI: 0.52, 0.76; $p < 0.0001$). The OS endpoint was not met; at the final analysis (FA) of OS, the median OS was 18.6 months in the Dato-DXd arm and 18.3 months in the ICC arm (HR: 1.01, 95% CI: 0.83, 1.22). Although there was no OS improvement, Dato-DXd was also not associated with a clear trend toward potential detriment compared to ICC. The most commonly reported adverse reactions ($\geq 20\%$) with Dato-DXd were stomatitis, nausea, fatigue, alopecia, constipation, dry eye, keratitis, and vomiting. Overall, the favorable benefit-risk profile for Dato-DXd supported its approval for the intended indication.

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Disclosure of Potential Conflicts of Interest: The authors report no financial interests or relationships with the commercial sponsors of any products discussed in this report.

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Paul G. Kluetz completed work on this article while employed at the FDA, and at the time of publication is an employee at Paradigm Health. Melissa Pegues completed work on this article while employed at the FDA, and at the time of publication is an employee at Salamandra, LLC.

Introduction

Breast cancer (BC) is the most common cancer in women worldwide (1). In the United States (US), there were 310,720 new cases and 42,250 deaths from BC estimated for 2024 (2). Although rare, male patients can also develop BC and may present at a higher stage than female patients. A total of 2,790 new cases and 530 deaths from BC were estimated for male patients in 2024 (2). Approximately 70% of patients with BC have HR+, HER2– disease (3). In the US, the preferred first-line treatment for patients with unresectable or metastatic HR+, HER2– BC is a cyclin dependent kinase 4/6 inhibitor (CDK 4/6i) in combination with endocrine therapy (ET) based on multiple pivotal trials demonstrating improvement in PFS (4, 5) and OS (6). At disease progression, patients can receive subsequent treatment with ET-based regimens (7 – 11). Once patients no longer benefit from ET-based therapy, they typically receive sequential single agent chemotherapy (7). More recently, patients with HR+, HER2-low (IHC 1+ or IHC 2+/ISH–) or HER2-ultralow (IHC 0 with membrane staining) BC have the option to receive fam-trastuzumab deruxtecan-nxki following disease progression on ET-based treatment in the metastatic setting (12). Patients with HR+, HER2-low tumors could also receive fam-trastuzumab deruxtecan-nxki following one line of chemotherapy in the metastatic setting or with disease recurrence during or within 6 months of completing adjuvant chemotherapy (13). Patients with HR+, HER2– tumors also have the option of receiving sacituzumab govitecan-hziy following ET and two additional prior lines of systemic therapy in the metastatic setting (14). However, metastatic HR+, HER2-negative BC remains incurable; therefore, better treatment options are needed. This article summarizes the data and FDA’s review leading to the approval of Dato-DXd.

Chemistry, Manufacturing, and Control

Dato-DXd is an antibody-drug conjugate (ADC) composed of a humanized IgG1 anti-trophoblast cell-surface antigen 2 (Trop2) monoclonal antibody covalently linked via a protease cleavable linker to a topoisomerase I inhibitor, DXd, with a drug-to-antibody ratio of 4 (15). The antibody is produced using recombinant DNA technology in Chinese hamster ovary cells, and the topoisomerase inhibitor and linker are produced by chemical synthesis. The drug product is a sterile, white to yellowish white, preservative-free lyophilized powder in single-dose vials for reconstitution and further dilution.

Nonclinical Pharmacology and Toxicology

Dato-DXd binds to anti-trophoblast cell-surface antigen 2 (Trop2) on the surface of cells, is internalized, and proteolytic cleavage of the linker releases the cytotoxic payload (DXd). DXd, an exatecan derivative, inhibits topoisomerase I, resulting in DNA damage and apoptosis (16). Dato-DXd inhibited cell proliferation in Trop2-positive cancer cell lines (16) and showed anti-tumor activity in patient-derived breast cancer xenograft mouse models (17).

In repeat-dose toxicity studies, intravenous administration of Dato-DXd to rats and monkeys for up to 3 months resulted in toxicity in the gastrointestinal tract, lung, thymus, skin, eye, and kidney, which was reversible by the end of the recovery period. The toxicity profile of Dato-DXd in animals was consistent with adverse reactions (ARs) observed in clinical trials.

In rats, Dato-DXd caused degeneration and atrophy of testes, decreased number of sperm, and increased atretic follicles in the ovary that did not recover after a 2-month recovery period. Thus, Dato-DXd may result in impairment of fertility in males and females based on animal findings. DXd demonstrated clastogenic potential in an *in vitro* chromosome aberration assay and an *in vivo* bone marrow micronucleus assay. Based on the mechanism of action, the genotoxicity of DXd and its effects on actively dividing cells, and Trop2 expression in embryonic tissues and the placenta (15), Dato-DXd is expected to cause reproductive and embryofetal toxicity. Embryo-fetal toxicity is included as a Warning in Dato-DXd labeling (18).

Clinical Pharmacology

The recommended dosage of Dato-DXd is 6 mg/kg (up to 540 mg) administered as an IV infusion once every 3 weeks. The recommended dosage was supported by the totality of data including exposure-response (E-R) analyses for efficacy and safety in patients with BC, and the observed rates of dosage modifications due to treatment-emergent adverse events (TEAEs). For patients weighing ≥ 90 kg, a dose cap of 540 mg is recommended to reduce the risk of ocular surface toxicity and other ARs.

Intrinsic factors including age, sex, mild or moderate renal impairment (RI), mild hepatic impairment (HI), tumor type, and Trop2 expression are unlikely to have clinically meaningful effects on the exposure of Dato-DXd. No dosage adjustment is recommended based on intrinsic factors other than body weight or for coadministration with CYP3A or OATP1B inhibitors. Patients with mild or moderate RI have an increased risk of interstitial lung disease (ILD) compared to patients without RI and should be monitored for increased respiratory ARs. Limited pharmacokinetic (PK) data from patients with moderate HI showed increased DXd exposure compared to patients without HI. Based on the available PK data and positive E-R relationships for safety, patients with moderate HI should be monitored for increased ARs.

The anti-drug antibody (ADA) assay was inadequate to detect ADA in the presence of Dato-DXd at clinically relevant concentrations which may interfere with the characterization of ADA in clinical samples (i.e., false-negatives). A postmarketing commitment (PMC) was therefore issued to complete high-resolution drug tolerance characterization for the ADA assay.

Clinical Trial Design:

The efficacy and safety of Dato-DXd was evaluated in a multicenter, open-label, randomized (1:1) trial of Dato-DXd compared to the investigator's choice chemotherapy (ICC) in patients with inoperable or metastatic HR+, HER2- BC who had been treated with one or two prior lines of systemic chemotherapy in the inoperable or metastatic setting (Figure 1). Selection of ICC occurred prior to randomization. Randomization was stratified by the number of previous lines of chemotherapy, geographic region, and prior use of CDK4/6i. Treatment was given until disease progression or unacceptable toxicity. No crossover was permitted at disease progression. The dual primary endpoints were PFS assessed by blinded independent central review (BICR) according to Response Evaluation Criteria in Solid

Tumors (RECIST) v1.1 and OS. One analysis for PFS, and two interim analyses (IA) and a final analysis (FA) for OS were planned. The confirmed overall response rate (ORR) and duration of response (DOR) assessed by BICR according to RECIST v1.1 were secondary endpoints which were not part of the statistical hierarchy. To control type I error rate at a two-sided alpha level of 5%, an alpha level of 1% was allocated to the PFS analysis and 4% to the OS analysis. As the PFS results crossed the efficacy boundary, the 1% alpha allocated to the PFS analysis was re-allocated to the OS analysis so that OS was analyzed at a 5% alpha level. Descriptive patient-reported outcomes were also collected.

Results:

A total of 732 patients with inoperable or metastatic HR+, HER2- BC comprised the intent-to-treat (ITT) population, with 365 in the Dato-DXd arm and 367 in the ICC arm. The most common chemotherapy in the ICC arm was eribulin (60%), followed by capecitabine (21%), vinorelbine (10%), and gemcitabine (9%). Most variables were well-balanced between the arms, with no imbalance likely to have affected the results. Key patient demographics and baseline disease characteristics are shown in Table 1.

Efficacy:

TB01 met its primary endpoint demonstrating a statistically significant improvement in PFS (HR=0.63 [95% CI: 0.52, 0.76]; median PFS: 6.9 vs. 4.9 months). Subgroup analyses showed consistency of PFS effect across important subgroups. Sensitivity analyses, including those assessing the impact of early censoring, and PFS per investigator results supported the primary PFS per BICR findings. Confirmed ORR was numerically higher in the Dato-DXd arm compared to the ICC arm (Table 2).

At time of PFS analysis, OS data were immature, and more mature OS data were submitted throughout FDA's review to support the benefit-risk assessment. At the final analysis, TB01 did not demonstrate benefit in OS (HR=1.01 [95% CI: 0.83, 1.22]). The observed final OS HR point estimate was approximately 1, and multiple sensitivity analyses were consistent with the primary OS findings. In additional exploratory analyses, the restricted mean of survival time (RMST) difference point estimates were positive favoring Dato-DXd across various cutoff timepoints. Based on these results, the FDA concluded that while Dato-DXd was not associated with an OS improvement, it also did not appear to have an apparent detrimental effect on OS.

Exploring efficacy (PFS, OS, and ORR) by Trop2 expression group showed no apparent association between outcomes and Trop2 expression groups. However, as Trop2 expression was not a randomization stratification factor, the results observed by Trop2 could be confounded by imbalanced baseline characteristics; therefore, the subgroup results should be interpreted with caution.

Safety:

The FDA's independent safety analysis was based on the 711 patients in the safety population of TB01 who received at least one dose of trial therapy, including 360 who received Dato-DXd and 351 who received ICC. Almost all patients experienced a treatment-

emergent adverse event (TEAE). The incidences of Grade 3–4 TEAEs were lower with Dato-DXd compared to ICC (33% vs. 54%), while the incidences of serious adverse events (SAEs) were similar (15% vs. 18%). TEAEs leading to dose reduction were also lower in patients who received Dato-DXd (23%) compared to ICC (32%); similarly for dosage interruptions (22% vs. 34%, respectively). Rates of treatment discontinuation due to TEAEs were ~3% in both arms.

The most common ($\geq 20\%$) TEAEs, including laboratory abnormalities, in patients treated with Dato-DXd were stomatitis, nausea, fatigue, decreased leukocytes, decreased calcium, alopecia, decreased lymphocytes, decreased hemoglobin, constipation, decreased neutrophils, dry eye, vomiting, increased ALT, keratitis, increased AST, and increased alkaline phosphatase. The most common ($\geq 5\%$) grade 3–4 TEAEs were decreased lymphocytes and stomatitis.

The FDA's review focused on understanding the toxicity of Dato-DXd compared to ICC. Important safety signals for Dato-DXd included ILD/pneumonitis, stomatitis, and ocular ARs; all of which are labeled as Warnings in the US drug labeling (18). ILD/pneumonitis occurred in 4.2% of patients treated with Dato-DXd vs. 0% with ICC. Although the reported incidence of ILD/pneumonitis in TB01 was low, prompt recognition and treatment are required to prevent life-threatening sequelae. Stomatitis occurred in 59% and 7% for all-grade and Grade 3 events, respectively, of patients who received Dato-DXd compared to 17% and 2.6% of patients who received ICC. An oral care plan including prophylactic steroid-containing mouthwash is recommended in labeling (18) to mitigate Dato-DXd-induced stomatitis. Ocular ARs including dry eye and keratitis (includes corneal disorder, erosion, infiltrates, lesion, toxicity, injury corneal; keratitis, keratopathy, punctate keratitis, and ulcerative keratitis) occurred in more patients who received Dato-DXd compared to ICC: 27% vs. 13% and 24% vs. 10%, respectively. A total of 51% of patients who received Dato-DXd experienced an ocular AR. The incidence of Grade 3 events was 1.9%, and no Grade 4 events occurred. Patients with clinically significant corneal disease were excluded from TB01. Ophthalmic exam at treatment initiation, annually while on treatment, at end of treatment, and if clinically indicated are recommended in labeling (18). In contrast, hematologic toxicities, both all grade and grade 3–4 were more notable in the ICC arm compared to Dato-DXd. Grade 3–4 neutropenia occurred in 35% of patients receiving ICC compared to 1.6% of patients receiving Dato-DXd.

Patient-reported symptom assessments included using PRO-CTCAE EORTC IL 117 (dry eye), and Patients' Global Impression of Treatment Tolerability (PGI-TT) items were collected weekly through week 12, then every 3 weeks thereafter until treatment discontinuation. Patients treated with Dato-DXd reported higher frequency, amount of, or severity for mouth sores, dry eye, and hair loss while patients in the ICC arm reported more hand-foot syndrome and neuropathy. In terms of overall side effect bother, patient reported bother was worse in patients receiving Dato-DXd compared to patients receiving ICC at each timepoint based on PGI-TT results. The patient-reported outcomes (PRO) data support the conclusion that Dato-DXd had more symptomatic AEs than ICC; however, we note that ICC was predominantly associated with hematologic toxicities which are typically asymptomatic.

Regulatory Insights

The FDA granted Dato-DXd traditional approval based on efficacy and safety results from TB01, a multicenter, open-label, randomized trial of 732 patients with unresectable or metastatic HR+/HER2- BC who had received prior endocrine-based therapy and 1 or 2 prior lines of chemotherapy in the unresectable or metastatic setting. TB01 met its PFS per BICR endpoint, and although the improvement in PFS was modest (~2 months), it was robust to multiple sensitivity analyses. Additionally, the ORR and DOR results supported the primary PFS per BICR findings. The OS HR was approximately 1, and despite the lack of improvement in OS, there did not appear to be a clear trend toward potential OS detriment.

The FDA examined the OS HR of approximately 1 and conducted additional sensitivity and RMST analyses using various cutoff timepoints. Results of OS sensitivity analyses were consistent with the primary OS findings, and RMST analyses favored Dato-DXd across all cutoff timepoints assessed, allowing the FDA to conclude that there was no clear detrimental effect on OS. The FDA considered these OS results acceptable within the context of a replacement trial in which the experimental arm does not have increased toxicity compared to the control arm.

TB01 had a replacement or “head-to-head” design, directly comparing Dato-DXd directly to ICC. In a “head-to-head” trial, a more modest PFS improvement can support a favorable regulatory decision, provided the standard-of-care control arm is active, because the PFS benefit in the experimental arm is in addition to the PFS benefit of the standard-of-care treatment. The PFS improvement should be robust across sensitivity analyses and supported by results from secondary endpoints. Furthermore, to support an approval, the experimental arm cannot introduce unacceptable toxicity compared to the control arm.

The FDA compared the safety and tolerability profile of Dato-DXd to ICC and determined that each therapeutic option had different side effect profiles. Common TEAEs associated with Dato-DXd, including nausea, stomatitis, and ocular toxicities, can be bothersome to patients even when low-grade. Patients may be symptomatic, require concomitant medications, and prolonged symptoms may impact quality of life. Results of PGI-TT, a PRO, suggested slightly worse levels of global impression of treatment tolerability for Dato-DXd compared to ICC. Dato-DXd was also associated with increased risk of ILD/pneumonitis. On the other hand, ICC was associated with increased hematologic toxicity compared to Dato-DXd, including increased Grade 3–4 neutropenia. Investigator and patient-reported data showed differing safety and tolerability profiles for Dato-DXd and ICC, and choice of therapy could be predicated on pre-existing conditions and avoidance of specific toxicities (e.g. myelosuppression).

Of note, some patients eligible for TB01 would now have additional standard-of-care treatment options. Patients who have HER2-low (IHC 1+ or 2+/ISH-) tumors (formerly included in the HER2-negative category) who have received one prior line of chemotherapy in the metastatic setting may be eligible for fam-trastuzumab deruxtecan-nxki, and patients who have received prior endocrine therapy and at least two additional lines of chemotherapy in the metastatic setting may be eligible for sacituzumab govitecan-hziy. The FDA does not have a comparative effectiveness requirement for a traditional approval, and therefore,

the experimental treatment does not need to show superiority to all available therapies. The trial supporting approval must be adequately designed with an acceptable control arm, and the FDA considered the TB01 control arm appropriate when the trial was initiated. Unfortunately, the activity of Dato-DXd following other approved ADCs and the activity of approved ADCs following Dato-DXd are unknown. Optimal sequencing of treatments, including ADCs, following ET is of paramount importance to patients with HR+, HER2– BC. Ongoing and future trials may eventually provide insight to these outstanding questions (19–21).

Conclusion

Results from TB-01 demonstrate a favorable benefit-risk profile (Table 3) for Dato-DXd. This approval represents another treatment option that is less myelosuppressive for patients with HR+, HER2– BC who require further treatment but are no longer candidates for further ET and have received prior chemotherapy in the unresectable or metastatic setting.

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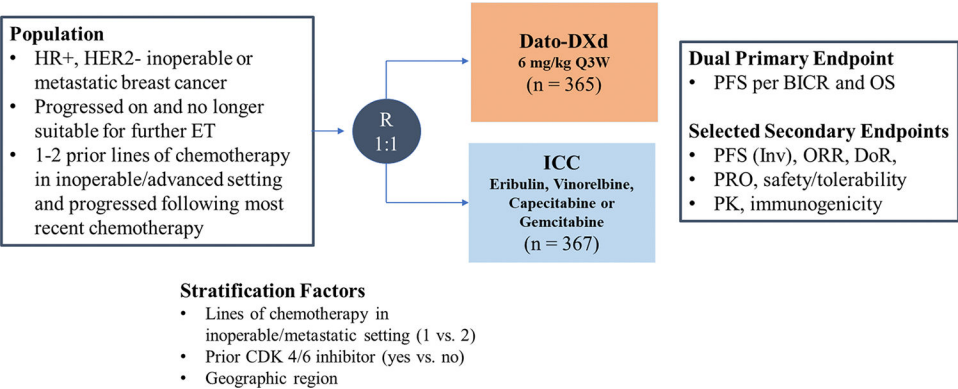


Figure 1: Trial Schema of TROPION-Breast01

Source: Dato-DXd Assessment Aid, Review of relevant individual trials used to support efficacy section; adapted from Applicant Figure 7.

Abbreviations: BICR- blinded independent central review; CDK- cyclin-dependent kinase; DoR- duration of response; ET – endocrine therapy; HER2– human epidermal growth factor receptor 2; HR– hormone receptor; ICC – investigator’s choice chemotherapy; inv- investigator; ORR- overall response rate; OS- overall survival; PFS- progression free survival; PK- pharmacokinetics; PRO- patient-reported outcome; Q3W- every 3 weeks;

Table 1:
Key patient demographics and baseline disease characteristics

Parameter, Statistic	Dato-DXd (N = 365)	ICC (N =367)	Total (N = 732)
Median age (range), years	56 (29, 86)	54 (28, 86)	55 (28, 86)
Age group (years), n (%)			
< 65	274 (75.1)	295 (80.4)	569 (77.7)
≥ 65	91 (24.9)	72 (19.6)	163 (22.3)
Sex, n (%)			
Female	360 (98.6)	363 (98.9)	723 (98.8)
Male	5 (1.4)	4 (1.1)	9 (1.2)
Race, n (%)			
White	180 (49.3)	170 (46.3)	350 (47.8)
Asian	146 (40.0)	152 (41.4)	298 (40.7)
Black or African American	4 (1.1)	7 (1.9)	11 (1.5)
Other	3 (0.8)	6 (1.6)	9 (1.2)
Not reported	32 (8.8)	32 (8.7)	64 (8.7)
Ethnicity, n (%)			
Hispanic or Latino	40 (11.0)	43 (11.7)	83 (11.3)
Not Hispanic or Latino	322 (88.2)	318 (86.6)	640 (87.4)
Missing	3 (0.8)	6 (1.6)	9 (1.2)
ECOG performance status, n (%)			
0	197 (54.0)	220 (59.9)	417 (57.0)
1	165 (45.2)	145 (39.5)	310 (42.3)
2	3 (0.8)	1 (0.3)	4 (0.5)
Previous lines of chemotherapy in the metastatic setting, n (%)			
1	229 (62.7)	225 (61.3)	454 (62.0)
2	135 (37.0)	141 (38.4)	276 (37.7)
≥3	1 (0.3)	1 (0.3)	2 (0.2)
Prior use of CDK 4/6 inhibitor, n (%)			
Yes	304 (83.3)	300 (81.7)	604 (82.5)
No	61 (16.7)	67 (18.3)	128 (17.5)
Geographic region			
US, Canada, Europe	186 (51.0)	182 (49.6)	368 (50.3)
Rest of World	179 (49.0)	185 (50.4)	364 (49.7)

Source: Dato-DXd Assessment Aid, Study results section; adapted from Applicant Table 20.

Table 2:**Efficacy Results**

	Dato-DXd (N = 365)	ICC (N =367)
Progression-Free Survival ^a		
Number of events (%)	212 (58)	235 (64)
Progressive Disease	201 (55)	218 (59)
Death	11 (3)	17 (5)
Median, months (95% CI)	6.9 (5.7, 7.4)	4.9 (4.2, 5.5)
Hazard ratio (95% CI) ^b	0.63 (0.52, 0.76)	
p-value ^{c, d}	<0.0001	
Overall Survival		
Number of events (%)	223 (61)	213 (58)
Median, months (95% CI)	18.6 (17.3, 20.1)	18.3 (17.3, 20.5)
Hazard ratio (95% CI) ^b	1.01 (0.83, 1.22)	
p-value ^c	NS	
Confirmed Objective Response Rate ^a		
n (%)	133 (36)	84 (23)
(95% CI)	(31, 42)	(19, 28)
Complete Response n (%)	2 (0.5)	0
Partial Response n (%)	131 (36)	84 (23)
Duration of Response ^a		
Median, months (95% CI)	6.7 (5.6, 9.8)	5.7 (4.9, 6.8)

Source: Dato-DXd Assessment Aid, Efficacy results section, adapted from FDA's analyses/assessment.

Abbreviations: CI- Confidence interval; NS- Not statistically significant

^a Assessed by BICR

^b Based on the stratified Cox proportional hazards model

^c Two-sided p-value based on stratified log-rank test

^d p-value is compared with the allocated alpha of 0.01

Table 3:**FDA Benefit-Risk Assessment of Dato-DXd**

Dimension	Evidence and Uncertainties	Conclusion and Reasons
Analysis of Condition	- Breast cancer is the most common cancer in women with 310,720 new cases and 42,250 deaths expected from breast cancer in the U.S. in 2024. - Although rare, breast cancer can also occur in male patients, with 2,790 cases and 530 deaths expected from breast cancer in the U.S. in 2024. - Approximately 70% of patients with breast cancer will have HR+, HER2- (IHC 0, 1+, or 2+/ISH-) disease.	Unresectable or metastatic HR+, HER2- breast cancer is a serious and life-threatening condition.
Current Treatment Options	- Unresectable or metastatic HR+, HER2- breast cancer is not curable. Treatment aims include reducing cancer-related symptoms, delaying disease progression, and prolonging survival. - Patients in the U.S. typically receive 1L treatment with a CDK4/6 inhibitor in combination with endocrine ET, followed by other ET-based regimens. - Once patients are no longer deriving benefit from ET-based regimens, they typically receive treatment with sequential single agent chemotherapies, such as capecitabine, eribulin, and other agents. - More recently, patients may have the option to receive ADCs such as T-DXd or SG. Neither T-DXd nor SG was approved when TB01 was initiated.	All treatment options are palliative. There is an unmet medical need to improve outcomes in patients with unresectable or metastatic HR+/HER2- breast cancer.
Benefit	- TB01 enrolled 732 patients with unresectable or metastatic HR+/HER2- breast cancer who had progressed on endocrine therapy and had received 1 or 2 prior lines of chemotherapy in the unresectable or metastatic setting. - Median PFS per BICR was 6.9 months in the Dato-DXd arm compared with 4.9 months in the ICC arm (HR 0.63, 95% CI 0.52–0.76, $p < 0.0001$). - The OS endpoint was not met. Median OS was 18.6 months in Dato-DXd arm and 18.3 months in the ICC arm, and the OS HR was 1.01 (95% CI 0.83, 1.22) indicating no trend towards benefit, but also no clear trend towards potential detriment. - Confirmed ORR was numerically higher in the Dato-DXd arm compared to the ICC arm: 36% (95% CI: 31, 42) vs. 23% (95% CI: 19, 28). Median DOR was 6.7 months (95% CI: 5.6, 9.8) in the Dato-DXd arm compared to 5.7 months (95% CI: 4.9, 6.8) in the ICC arm.	In a head-to-head trial, Dato-DXd was associated with a statistically significant improvement in PFS and an OS HR~1, with no clear trend toward potential detriment but also no improvement. Dato-DXd is another treatment option for patients with unresectable or metastatic HR+/HER2- breast cancer.
Risk and Risk Management	- Fatal TEAEs, SAEs, and TEAEs leading to drug discontinuations were similar in patients who received Dato-DXd compared to ICC, occurring in 0.3% vs. 0.9%, 15% vs. 18%, and 3.1% vs. 2.8% of patients, respectively. - Grade 3–4 TEAEs were lower in patients who received Dato-DXd (33%) compared to ICC (54%). Dose reductions and dosage interruptions were also lower with Dato-DXd compared to ICC: 23% vs. 32%, and 22% vs. 34%, respectively. This was partially due to decreased hematologic toxicity associated with Dato-DXd compared to ICC. - Dato-DXd was associated with increased all-grade, and grade 3 stomatitis compared to ICC: 59% vs. 17% and 7% vs. 2.6%, respectively. Compared to ICC, Dato-DXd was also associated with increased nausea: 56% vs. 27%, constipation: 34% vs. 17%, and vomiting: 24% vs. 12%. - There were increased ocular adverse reactions with Dato-DXd compared to ICC, including dry eye: 27% vs. 13% and keratitis: 24% vs. 10%. - ILD/pneumonitis is a serious risk associated with Dato-DXd, occurring in 4.2% of patients who received Dato-DXd and leading to death in 1 patient.	For patients with unresectable or metastatic HR+/HER2- breast cancer, Dato-DXd had an acceptable safety profile. Safe use of Dato-DXd can be managed with labeling. ILD/Pneumonitis, stomatitis, Ocular Adverse Reactions, and Embryo-Fetal Toxicity are included under Warnings.

Source: Dato-DXd Assessment Aid, FDA benefit-risk assessment section.

Abbreviations: ADC- antibody drug conjugate, BICR- blinded independent central review

CDK- cyclin dependent kinase, CI- confidence interval, Dato-DXd- datopotamab deruxtecan-dlnk, DOR- duration of response, ET- endocrine therapy, HER2- human epidermal growth factor receptor 2, HR- hazard ratio, HR+- hormone receptor-positive, ICC- investigator's choice chemotherapy, IHC- immunohistochemistry, ILD- interstitial lung disease, ISH- in situ hybridization, ORR- overall response rate, OS- overall survival, PFS- progression free survival, SAE- serious adverse event, SG- sacituzumab govitecan-hziy, TB01- TROPION Breast01, T-DXd- fam-trastuzumab deruxtecan-nxki, TEAE- treatment emergent adverse event, U.S.- United States